

REVIEW

How to measure response diversity

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Abstract

1. The insurance effect of biodiversity—that diversity stabilises aggregate ecosystem properties—is mechanistically underlain by inter- and intraspecific trait variation in organismal responses to the environment. This variation, termed *response diversity*, is therefore a potentially critical determinant of ecological stability. However, response diversity has yet to be widely quantified, possibly due to difficulties in its measurement. Even when it has been measured, approaches have varied.
2. Here, we review methods for measuring response diversity and from them distil a methodological framework for quantifying response diversity from experimental and/or observational data, which can be practically applied in laboratory and field settings across a range of taxa.
3. Previous empirical studies on response diversity most commonly invoke response traits as proxies aimed at capturing species' ecological responses to the environment. Our approach, which is based on environment-dependent ecological responses to any biotic or abiotic environmental variable, is conceptually simple and robust to any form of environmental response, including nonlinear responses. Given its derivation from empirical data on species' ecological responses, this approach should more directly reflect response diversity than the trait-based approach dominant in the literature.
4. By capturing even subtle inter- or intraspecific variation in environmental responses, and environment dependencies in response diversity, we hope this framework will motivate tests of the diversity–stability relationship from a new perspective, and provide an approach for mapping, monitoring and conserving this critical dimension of biodiversity.

KEYWORDS

diversity–stability relationship, functional diversity, functional trait, insurance hypothesis, response trait, species asynchrony, temporal variability

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1 | INTRODUCTION

Ecological stability has been a core focus of ecology since the 1950s when interest in the relationship between diversity and stability arose (McCann, 2000). Ecological stability is a multidimensional concept encompassing a variety of metrics for measuring resistance to and recovery from disturbance, and variability in time and space (Donohue et al., 2013; Pimm, 1984). Early observations suggested that more diverse communities are more stable in their ability to resist invasions or in the magnitude of their fluctuations (Elton, 1958; MacArthur, 1955; Odum, 1953). Using randomly structured model communities, May (1973) then demonstrated that diversity per se did not increase stability (i.e. did not dampen population fluctuations), but rather decreased it (see also Pimm & Lawton, 1978; Yodzis, 1981). Subsequent theoretical developments (e.g. Ives & Carpenter, 2007; Kondoh, 2003; McCann et al., 1998) reached the consensus that species richness tends to stabilise aggregate ecosystem properties when complexity (in terms of, e.g. variable environments, interaction strengths or behaviours) is higher (Ives et al., 1999; but see Jacquet et al., 2016; Pennekamp et al., 2018). This is because declines in one species' abundance or *performance* will be offset by neutral or positive responses of other species or phenotypes (reviewed by Loreau et al., 2021). We define performance as any measurable feature an organism or population that is an indicator of individual fitness. Functional traits that indicate performance could include both higher-level traits (such as per-capita growth rate) and lower-level traits (such as morphological traits) in the trait hierarchy (Agrawal et al., 2010). High-level traits are generally more likely to relate to environmental conditions than low-level traits (Violle et al., 2007). Together, these ideas encapsulate the insurance effect of biodiversity, which posits that biodiversity both enhances and stabilises ecosystem functioning (Yachi & Loreau, 1999).

However, biodiversity is only the 'passive recipient' through which underlying ecological mechanisms drive stability (McCann, 2000; Shade, 2017). Asynchronous or compensatory dynamics among species through time are now widely considered a key driver of the diversity–stability relationship (e.g. Craven et al., 2018; Sasaki et al., 2019; see also Loreau et al., 2021). Many specific biological mechanisms have been proposed as drivers of stability—largely through their effect on species asynchrony—including: variable interaction strengths (e.g. McCann et al., 1998; Yodzis, 1981) or mixtures of interaction types (e.g. Hammill et al., 2015; Mougi & Kondoh, 2012); food web nestedness (e.g. Thebault & Fontaine, 2010); trophic flexibility and food web rewiring (e.g. Kondoh, 2003); dominant, rare or otherwise key species (e.g. Arnoldi et al., 2019; Ross, García Molinos, et al., 2022a; Sasaki & Lauenroth, 2011; Walker et al., 1999); and asynchronous patch dynamics in space (e.g. Loreau et al., 2003; Wang & Loreau, 2016). Recently, the concept of *response diversity* has also emerged as a potentially key, yet under-explored driver of stability (Kahiluoto et al., 2019; Pennekamp et al., 2018; Sasaki et al., 2019).

Response diversity characterises the range of responses to the environment displayed among members (e.g. individuals, species) of a focal group such as a population, guild or community and

is a priori predicted to enhance community (aggregate) stability (Elmqvist et al., 2003; Mori et al., 2013; Nyström, 2006). For example, Ives et al. (1999) posited that differences in how species respond to environmental fluctuations (i.e. response diversity) may explain species diversity's stabilising effect on total community biomass. In a later study, Ives and Carpenter (2007) found that, in a model of randomly structured competitive communities, response diversity's stabilising effect on communities negated the destabilising effects of interspecific interactions. The authors concluded that species–environment relationships were more important for stability than were interspecific interactions (Ives & Carpenter, 2007; see also Houlahan et al., 2007). Although such studies typically focus on species responses to fluctuating environments (e.g. Ives et al., 1999; Thibaut & Connolly, 2013), species' environmental responses are most easily characterised along environmental gradients (see Section 3). The importance of this distinction may be a matter of scale. The rate at which organisms experience environmental change impacts their physiological and ecological capacity to respond to such change (Pineck et al., 2020; Wolkovich et al., 2014), including the likelihood of local adaptation along a gradient and under temporally variable environmental conditions (e.g. Villa Martín et al., 2019). The perceived rate of environmental change is itself a consequence of life history: what one short-lived organism perceives as a gradual (or abrupt) environmental change may simultaneously be perceived as a temporary environmental flux by another, longer-lived organism (Jackson et al., 2021; Wolkovich et al., 2014). Measuring species–environment relationships over environmental gradients is therefore practically useful for informing about species and community responses to temporal environmental fluctuations.

Interspecific variation in species–environment relationships is critical for the maintenance of diversity in variable environments (Chesson, 2000) and species-rich systems are thus anticipated to exhibit a wider diversity of species–environment interactions compared to species-poor systems (Vogel et al., 2019). This response diversity may, in turn, support a variety of stabilising mechanisms acting on the aggregate properties of communities and ecosystems. For example, response diversity may be one candidate ecological mechanism underpinning many of the statistical explanations of the diversity–stability relationship, such as the portfolio effect (Loreau, 2010; Loreau et al., 2021; Tilman et al., 1998), particularly through its effect on species asynchrony. Response diversity can promote asynchrony in species' biomass or abundance through time (Sasaki et al., 2019) and species-specific tolerance to environmental conditions may be partly responsible for temporal and functional complementarity among species (Loreau & de Mazancourt, 2013; Petchey, 2003). Indeed, asynchrony may frequently be an outcome of response diversity (e.g. Sasaki et al., 2019), suggesting that the often-observed stabilising effect of species asynchrony on aggregate community or ecosystem properties (e.g. Hector et al., 2015; Morin et al., 2014), may result from underlying differences in species–environment relationships. It should therefore benefit those interested in the diversity–stability relationship to measure both species asynchrony and response diversity—given that response diversity

should predict asynchrony, and in turn stability—as well as any additional factors expected to drive community stability that may be of interest, such as species richness per se, spatial synchrony or population stability.

2 | EXISTING RESPONSE DIVERSITY STUDIES

Few empirical studies have explicitly measured response diversity. Recognising that many studies may measure response diversity without labelling it such, a systematic review of the literature returned 46 ecology papers that empirically measured response diversity (Supporting Information Figure S1). These empirical studies used various methods to measure response diversity. The majority ($n = 28$ studies, ~60%) used low-level functional traits (see Box 1 for discussion), where the diversity of response traits—that is, those traits that predict some aspect of how a species responds to the environment (Suding et al., 2008)—represents response diversity, measured using the functional dispersion [FDis] index (e.g. Hordley et al., 2021; Schnabel et al., 2021; Spasojevic et al., 2016). Some studies measuring the dispersion of response traits explicitly do so after defining functional effect groups (Laliberté et al., 2010; Morel et al., 2020; Sasaki et al., 2019; Thornhill et al., 2018) following Elmqvist et al. (2003), although subsequent definitions relax this requirement (e.g. Mori et al., 2013). Most empirical studies measuring response trait diversity do so for trees (e.g. Altomare et al., 2021; Craven et al., 2016; Schnabel et al., 2021) and other groups of plants (e.g. Döbert et al., 2017; Mandle & Ticktin, 2015; Morel et al., 2020), or for freshwater or terrestrial invertebrates (e.g. Mumme et al., 2015; Thornhill et al., 2018).

Another approach to measuring response diversity empirically is to define a species-specific interaction term in multispecies models of, for example, abundance as a function of the environment ($n = 7$ studies; Figure S1). This binary approach asks whether there is response diversity among species or not. If the species-specific interaction term passes some significance or model selection threshold, then species are claimed to differ in their abundance ~ environment slopes; that is, they exhibit response diversity (Winfree & Kremen, 2009). This approach to measuring response diversity has emerged primarily from the pollination literature (Bartomeus et al., 2013; Cariveau et al., 2013; Staver et al., 2017; Winfree & Kremen, 2009), but has been used elsewhere (Fauchald et al., 2011; Malyshev et al., 2016). Another related but less common approach to measuring response diversity is to measure some aspect of a population's performance, such as its intrinsic rate of increase (e.g. McCann, 2016) or biomass change in response to the environment, and take the range of performance–environment model slopes as a direct estimate of response diversity. Leary and Petchey (2009) did this with four ciliate species by modelling their intrinsic rates of increase against temperature when in isolation, first determining species-specific environmental responses. They then assembled experimental communities with different pairwise combinations of

these species, producing different levels of response diversity. A conceptually similar approach from Thibaut et al. (2012) used observational time series of reef fish abundance to measure response diversity based on the average correlation coefficient of random effects (intercept terms) on the intrinsic rates of increase between functional groups. Unlike Leary and Petchey's (2009) approach for considering pairwise performance–environment correlations, this approach directly quantifies covariation among responses to environmental fluctuations using pairwise correlations in the deviation from the average intrinsic rates (i.e. the random effect term; Thibaut et al., 2012). These various methods for measuring response diversity (Figure 1 and Figure S1) have emerged independently, resulting in a disconnect among empirical response diversity studies, and a general methodological framework for measuring response diversity is lacking.

In response to this, in the remainder of this article we present a framework for measuring response diversity empirically using performance–environment relationships (Section 3), including application to simulated data for linear (Section 3.1) and nonlinear species responses where the form of the performance–environment relationship depends on environmental context (Section 3.2), as well as an application to an empirical dataset of multispecies aquatic microcosms (Section 3.3). Finally, we discuss future directions for empirical studies of response diversity and the challenges of applying this approach in a range of contexts (Section 4).

3 | MEASURING RESPONSE DIVERSITY EMPIRICALLY

Here, we propose a framework for empirically measuring response diversity using a conceptually simple and empirically tractable approach that can be applied in observational and experimental settings in both the laboratory and field. We extend the models describing performance–environment relationships presented above to include any response variable that represents a species' ecological response along a continuous gradient of abiotic or biotic environmental condition. In contrast to most previous studies of response diversity which focus on 'low-level' functional traits such as nitrogen content or specific leaf area (Box 1), our approach considers response diversity a 'high-level' trait of per capita growth rates for instance, tying it more directly to species performance (see Agrawal et al., 2010; Violle et al., 2007). That said, low-level traits can still be used in this framework, and can in some cases be informative and important responses (e.g. stand-level nitrogen content).

Our aim is not to define a single best metric of response diversity for use in all cases, but rather to illustrate how it can be measured. We demonstrate two complimentary metrics with desirable properties for our purposes. More suitable metrics for different use-cases likely exist, and we encourage researchers to explore various methods while warning against the proliferation of untested response diversity metrics—a new metric should necessarily be an improvement on existing metrics in its ability to fulfil its purpose

BOX 1 Response trait diversity: a questionable indicator of stability?

Functional diversity measures the range and value of species' traits which, in turn, aim to represent those species' contributions ecosystem functioning (Tilman, 2001). Myriad functional diversity metrics are now used to estimate ecosystem functioning using functional traits (Cadotte et al., 2011; Díaz & Cabido, 2001; Petchey & Gaston, 2006). Given their effect on ecosystem processes and functions, such traits are now termed *effect* traits, while *response* traits aim to capture information about how a species responds to environmental change (Oliver et al., 2015; Suding et al., 2008). Response traits are often assumed to, rather than empirically demonstrated to link to species' environmental responses (Figure 1), and even where empirically tested, environmental responses are not always well captured by putative functional traits (Bartomeus et al., 2017). Examples of response traits include life-form and seed mass in plants (Sasaki et al., 2019), aquatic stage and reproductive strategy in freshwater macroinvertebrates (Thornhill et al., 2018), and clutch size and thermal maximum in birds (Hordley et al., 2021). Response trait diversity is then typically measured as the functional dispersion (FDis) of the response trait space (Laliberté et al., 2010; Rader et al., 2013), or sometimes using the closely related Rao's quadratic entropy (RaoQ) index (Chillo et al., 2011; Correia et al., 2018). Some studies alternatively group species into response groups (e.g. Joseph et al., 2014; Schweiger et al., 2015), where species within a functional group are posited to respond similarly to environmental change based on their similar response traits. Studies aiming to quantify functional redundancy have therefore typically relied on functional grouping (e.g. Bruno et al., 2016; Laliberté et al., 2010). However, this approach ignores a suite of functional differences between species; functional grouping is rarely ecologically meaningful (Petchey & Gaston, 2006).

Considerable challenges exist when quantifying functional diversity from traits and linking traits to ecological stability. For example, there is still no consensus on which traits are 'functional' in a range of systems (Violle et al., 2007), and many traits can be highly correlated with, for example, body size (Kozłowski & Weiner, 1997). When attempting to measure response trait diversity, it can be difficult to delineate responses from effects; many traits, such as specific leaf area or photosynthetic pathway (Sasaki et al., 2019), act both as response and effect traits depending on environmental context (Hooper et al., 2005; Suding et al., 2008). This difficulty in separating response and effect traits could be a critical flaw in many trait diversity studies given that the overlap between the identity of response and effect traits is expected to drive changes in ecosystem functioning with the environment (Suding & Goldstein, 2008); as the overlap between response and effect traits increases, changes in the environment more likely modify ecosystem functioning. Accordingly, ecosystem functioning is expected to be stable only in cases where effect traits and response traits do not overlap, else changes in response diversity will necessarily propagate to affect ecosystem functioning (de Bello et al., 2021).

It is also often practically challenging to measure functional traits in the field, particularly for mobile organisms (Moretti et al., 2017; Pakeman & Queded, 2007). When measuring functional diversity from species traits, there is no single best metric for all scenarios, as most functional diversity indices capture distinct aspects of the functional differences within a community (Schleuter et al., 2010; Villéger et al., 2008). Linking trait diversity and stability through species asynchrony requires examining multiple response traits simultaneously (van Klink et al., 2019), which involves considerable sampling effort, as well as categorisation of traits as response and/or effect traits. Groups of species with similar response trait values should exhibit more synchronous population dynamics, while those with more distinct response traits are more likely to fluctuate asynchronously (Adler et al., 2013; Lepš et al., 2018). As communities are exposed to a wider range of environmental conditions, the importance of multi-trait dissimilarity in driving species asynchrony increases. Conversely, when interested in a specific environmental perturbation, individual response trait values may be more important for species' performance and ecosystem functioning than multidimensional response trait diversity (de Bello et al., 2021). Another metric of interest may be community-weighted mean (CWM) trait values, particularly when considering the effect of dominant species and their traits on stability (de Bello et al., 2021; Sasaki & Lauenroth, 2011).

Further methodological considerations include whether and how to abundance-weight functional diversity indices to account for rare species (Lavorel et al., 2008), and how to handle intraspecific trait variation given its ecological importance (de Bello et al., 2011; Herrando-Pérez et al., 2019; Ross et al., 2017). Another important consideration if linking trait diversity and stability is *when* to measure response trait diversity. Most studies linking traits and stability consider the response trait diversity of some average community through time (e.g. Craven et al., 2018), but this ignores temporal community turnover, thereby weakening the expected relationship between response trait diversity and stability (de Bello et al., 2021). Together, these challenges and others considerably limit the efficacy of using low-level functional traits to empirically quantify response diversity and measure its relationship with temporal stability. Unless using well-defined and empirically tested response traits (e.g. Kandlikar et al., 2022), we therefore suggest caution in measuring response diversity from low-level functional traits, and in assuming that such trait diversity should reflect any meaningful facet of ecological stability.

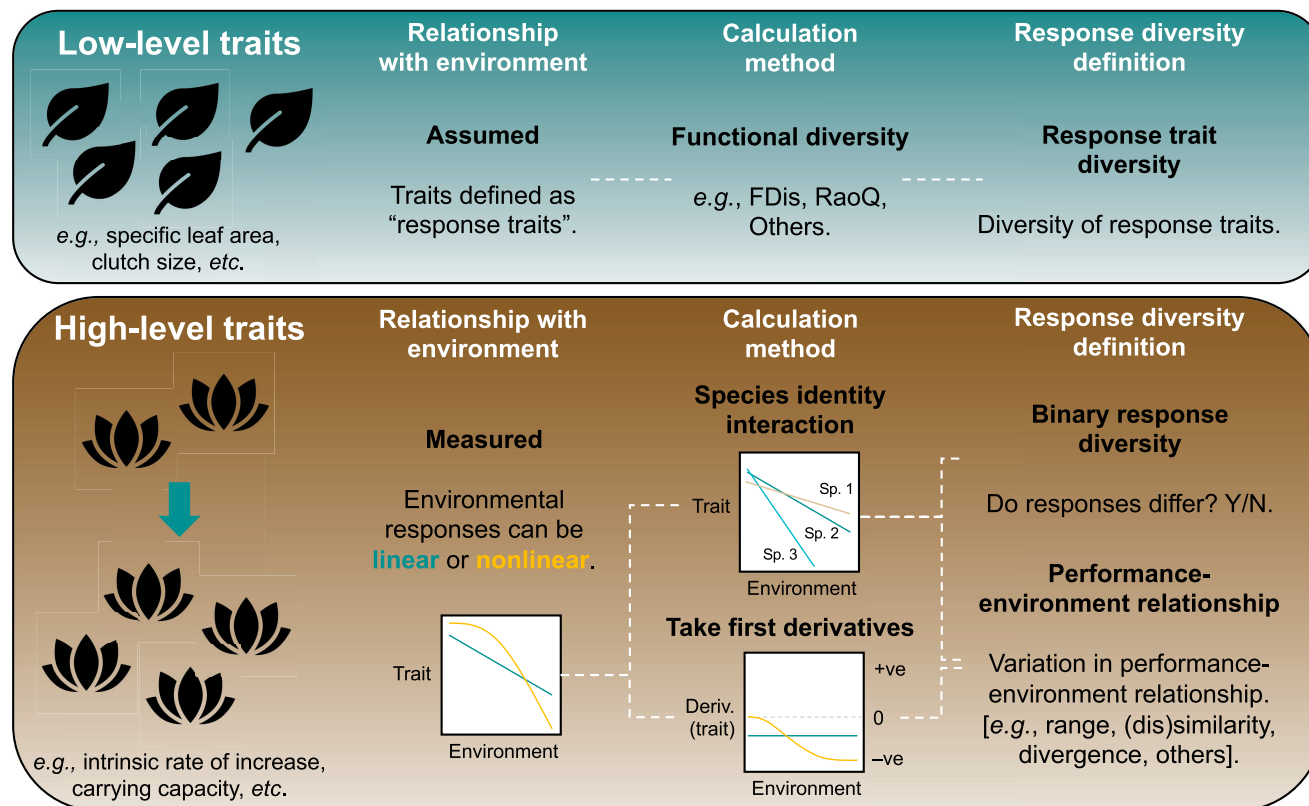


FIGURE 1 How to measure response diversity. Response diversity can be measured from low-level functional traits—the approach dominant in the literature (Figure S1)—or from higher-level traits which are themselves sometimes underlain by low-level traits. One limitation of measuring response diversity based on low-level traits is that such approaches often assume relationships between traits and the environment, rather than directly measuring trait–environment relationships (Box 1). Some previous studies have directly measured trait–environment relationships and tested whether different species respond differently to the environment (‘species identity interaction’; Section 2). Such measurements can be linear or nonlinear. We propose taking the first derivative of such trait–environment relationships to better account for nonlinear environmental responses (Section 3.2). One can then determine whether all species respond similarly to the environment (‘binary response diversity’; Section 2) or can go further and measure aspects of interspecific variation in trait–environment relationships (i.e. the ‘performance–environment relationship’; Section 3) using methods described here or elsewhere.

(see Figure 5 below). In response diversity's case, this purpose is to predict ecological stability, as grounded in the basic principles of the insurance hypothesis (Yachi & Loreau, 1999), while noting that response diversity may not always be the only driver of stability or species asynchrony.

We propose extending the performance–environment models used elsewhere (Leary & Petchey, 2009) into a generalisable framework for measuring response diversity. Despite rarely being used to date, this method is easily tractable in experimental and observational studies, and provides a more direct estimate of how species respond to the environment than, for example, the often-used response trait approach (Box 1). Given our definition of *performance*, the response variable in such models need not be intrinsic rate of increase, but could in fact be any relevant functional variable, such as biomass, abundance or an ecosystem function to which a species contributes. Similarly, the *environment* to which species are responding does not have to be an abiotic feature such as temperature; it can be any continuous biotic or abiotic condition. For example, one could model the responses along a land-use intensification gradient (Moore

& Olden, 2017; Stavert et al., 2017; Winfree & Kremen, 2009), or in response to the abundance of a predator, competitor or the availability of a prey species (e.g. Fauchald et al., 2011). The latter approach, using biotic conditions as the environment to which a species responds, is conceptually similar to modelling the functional response (Holling, 1966), which in turn may scale with abiotic conditions such as temperature (Daugaard et al., 2019). Likewise, modelling biomass change in response to a competitor is akin to concepts in coexistence theory such as negative density dependence (e.g. Armitage & Jones, 2019) or invasion growth rate (Grainger et al., 2019).

3.1 | Measuring variation among performance curves

From such models, we suggest measuring response diversity as the variation in the slope (first derivative) of a performance–environment function evaluated over the environmental gradient in question. Specifically, we suggest using generalised additive models (GAMs) to

fit performance–environment relationships individually for each species using smoothing parameters as appropriate (Figures 2a,c), then taking the first derivatives of these models to estimate model slopes along the environmental axis (Figures 2d–f). Note that it may not always be necessary to calculate the derivatives of a performance

variable that is already a rate measure; in this case, the fitted GAM function would suffice. GAMs are free from many of the assumptions of linear regressions, such as normality and homoscedasticity of residuals, and can be used to evaluate both linear and nonlinear responses, or a mixture of the two (Hastie & Tibshirani, 1987; see Ross,

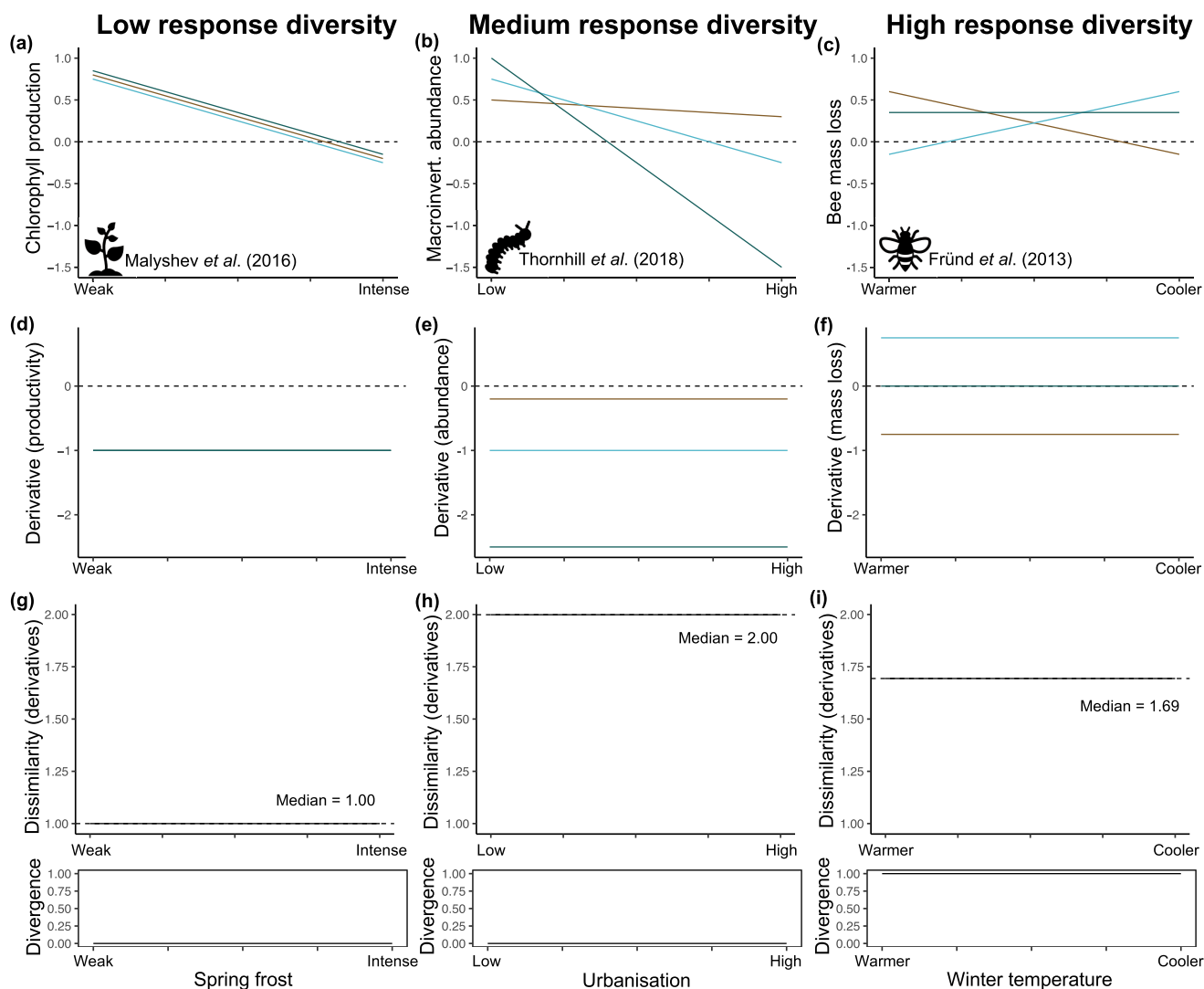


FIGURE 2 Measuring response diversity for linear species–environment relationships. We reimaged the original analyses of three studies, using standardised, simulated data to demonstrate how response diversity could be measured practically under our framework. (a) We proposed a hypothetical case (based on Malyshev et al., 2016), where *Chlorophyll* production (which is often an indicator of biomass, particularly for aquatic plants) declines linearly across several species [coloured lines] of plants with spring frost intensity, with no difference among species in the slope of this decline. (d) We calculated the first derivatives of these responses, which in this case was identical for all species. (g) Then, we measured response diversity using the proposed similarity-based diversity metric (see main text), and found it to be one (its lower bound), since all species respond identically to spring frost. Our divergence metric was zero in this case, as the first derivatives do not span zero. (b) Next consider a case (based on Thornhill et al., 2018), where the abundance of several species of macroinvertebrates declines linearly, but at different magnitudes, along an urbanisation gradient. (e) Here, the first derivatives differed (owing to differing model slopes), resulting in higher dissimilarity (h). Again, the derivatives did not span zero, so divergence was zero. (c) Finally, we might be interested in the decline in mass of several bee species as a function of winter temperature (see Fründ et al., 2013). In this example, species responses to winter temperature diverged (one species increased, one decreased, one was unaffected), but all such responses were linear, and the magnitude of responses was not as extreme as in (b). (f) Thus, the first derivatives were less spread out than in (e), but spanned zero. (i) Accordingly, dissimilarity was lower than for (h), but in this case, divergence was at its maximum value, 1, across the entire range of winter temperature values (the derivatives spanned zero at any temperature). Dashed lines indicate zero in panels a–f, and the median value of the similarity-based diversity metric (dissimilarity) along the X-axis in panels g–i (here, equivalent to any value along the X-axis since response diversity is constant).

Petchey, et al., 2022b). Response diversity can then be measured as the variation in the distribution of these derivatives to capture the variation in species-specific environmental responses (Figure 2g–i). In practice, this variation could be described using any suitable estimator (Figure 1), but here we use two complementary metrics.

First, we use a similarity-scaled metric of diversity introduced by Leinster and Cobbold (2012). Briefly, similarity-based diversity (Equation 1 in Leinster & Cobbold, 2012) is calculated based on pairwise Euclidean distances (i.e. *dissimilarity*) in performance–environment relationships between all pairs of species in the community. The metric is well suited to measuring response diversity, since it combines similarities and relative abundances, with an exponent (q) tuning the weighting placed on relative abundance (Leinster & Cobbold, 2012). Here, we fix q at zero, to focus solely on species richness (S), although this (dis)similarity metric can use information on species relative abundances within an $S \times S$ similarity matrix. We chose this estimator since it has several desirable properties for describing response diversity: it cannot decrease with species additions; does not increase with additions of redundant species responses; and captures even subtle differences in the distribution of different responses within the overall range. Dissimilarity should therefore be lowest (1) when all species respond identically (Figure 2g), and highest (S) when species responses are as dissimilar as possible (the estimator converges to species richness as performance–environment slopes diverge to infinity and $-\infty$).

Second, we develop a simple measure that accounts for whether performance–environment responses *diverge* in direction, since this should be a key indicator of the stabilising potential of response diversity under a given environmental condition. The insurance effect of biodiversity assumes that declines of one species can be offset by neutral or positive responses of others (Yachi & Loreau, 1999). As such, we also isolate and measure this property of response diversity using the formula:

$$\text{Divergence} = \frac{(\max[f'(E)] - \min[f'(E)]) - \left| \max[f'(E)] - \min[f'(E)] \right|}{(\max[f'(E)] - \min[f'(E)])},$$

where $f'(E)$ represents the first derivative of a performance–environment relationship, $f(E)$, at a particular environmental state (E). If the range of derivatives does not span zero (e.g. Figure 2d,e), then all species respond in the same direction to the environment (albeit with varying model slopes). In such cases, there is no response diversity along this divergence axis of species responses (the value is zero). Without interspecific interactions, even large differences in the magnitude of response (dissimilarity) cannot be stabilising (Ives et al., 1999; Mougi & Kondoh, 2012). In contrast, when derivatives are perfectly symmetric around zero, the value is one. When the range of first derivatives spans zero for a given environmental state (e.g. Figure 2f), species responses are diverging (at least one species is decreasing while another is increasing), which we expect a priori to be stabilising. This divergence metric is largest (1) when $\min[f'(E)] = -\max[f'(E)]$. Observation of stable community biomass or other aggregate ecosystem properties when the range of performance–environment

derivatives does not span zero would suggest that divergence in species responses is not the mechanism driving stability in that case.

Together, the similarity-based diversity metric and divergence metric make up two complementary components of response diversity; the similarity-based metric captures the overall magnitude of response diversity among species' responses (Leinster & Cobbold, 2012), while the divergence metric aims to quantify the extent to which species declines are offset by gains (Yachi & Loreau, 1999). Although often independent, these two metrics may respond similarly to changes in species' performance–environment relationships. For example, when mean responses (derivatives) are close to zero, an increase in dissimilarity will necessarily cause an increase in divergence. The specific metrics we use here may not be ideal for all use cases. Further work should consequently consider the need to develop suitable metrics of response diversity when those proposed fall short. There may also be opportunity to unite the two dimensions of response diversity discussed here into a single metric. For example, when (dis)similarity and divergence are highly correlated—such as when performance–environment relationships are nonlinear and divergence is high (Figure 3)—the simultaneous use of both metrics does not provide much additional information on response diversity over one metric in isolation (see Donohue et al., 2013), so using a single metric may suffice. That said, we demonstrate here the utility of these metrics for quantifying response diversity in a range of contexts (Figures 2 and 3), including applications to real data (Figures 4 and 5).

3.2 | Nonlinear responses to environmental variation

Ecological responses to the environment are often nonlinear, necessitating a wide range of statistical tools for considering such responses. When species' responses are nonlinear (e.g. Leary & Petchey, 2009), quantifying response diversity is not as simple as measuring the range of model slopes. However, the statistical framework we propose for measuring response diversity (i.e. measuring variation in performance–environment derivatives) is robust to nonlinearities such as species' responses to competition, predations or resource availability (Figure 3) or when tipping points are detected in empirical data (Berdugo et al., 2020). Figure 3 illustrates three such cases: one where the intrinsic rate of increase in flowering plant species responds nonlinearly but with low response diversity to total flowering plant abundance (Figure 3a); one where seedling establishment responds to grazing pressure with higher dissimilarity among species responses (Figure 3b); and one where trees respond nonlinearly to fire frequency, and diverge in their responses (Figure 3c). Our framework uses GAMs as they are robust even to cases where the form of the performance–environment relationship differs among species (Hastie & Tibshirani, 1987; Figure 3f,i). Where responses are nonlinear, the derivatives of performance–environment responses vary as a function of the environmental variable (along the X-axis in Figures 3d–f). In turn, this produces estimates of dissimilarity and

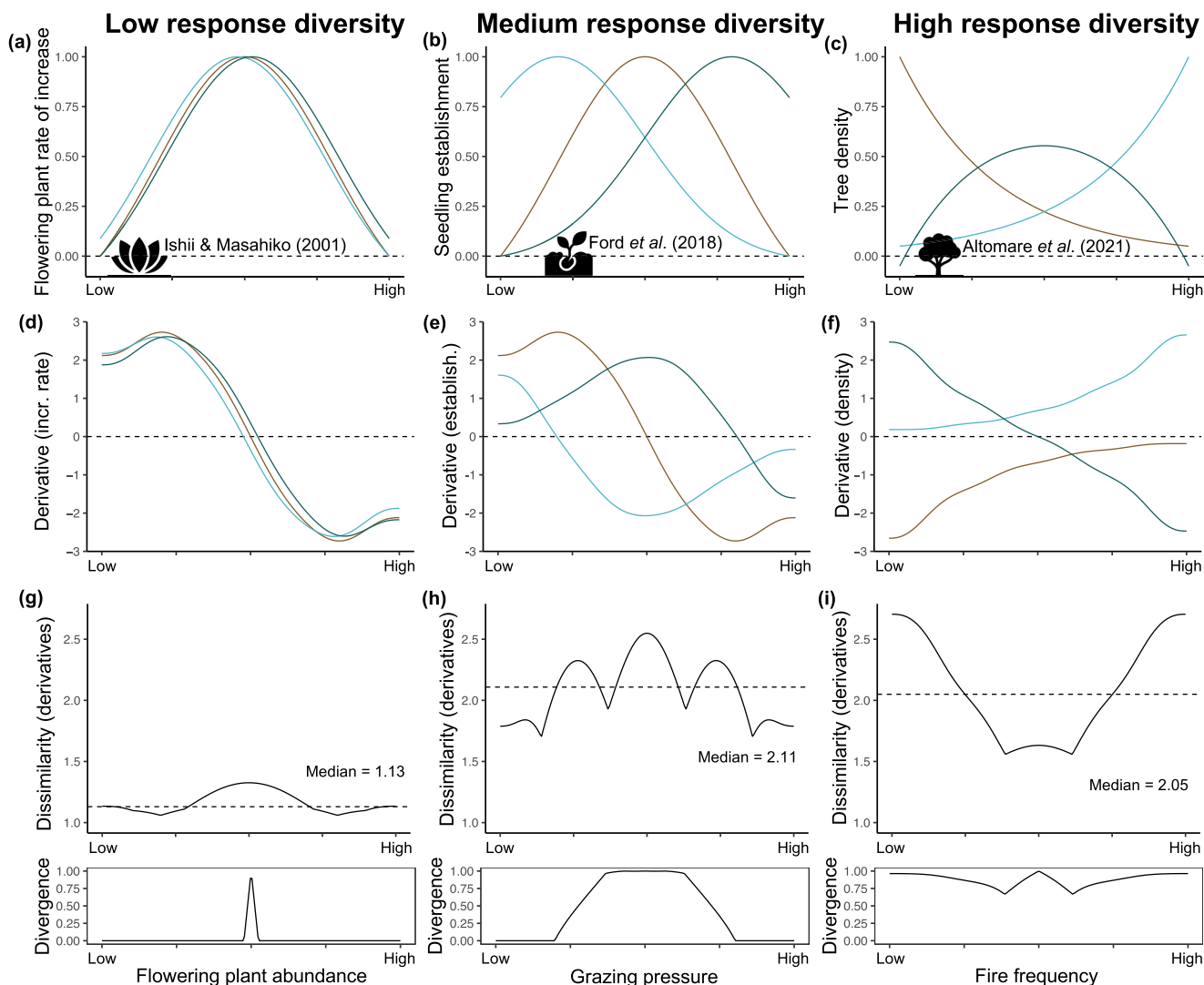


FIGURE 3 Measuring response diversity for nonlinear species–environment relationships. We reimaged the original analyses of three studies, using standardised, simulated data to demonstrate how response diversity could be measured practically under our framework. (a) We proposed a hypothetical case (based on Ishii & Masahiko, 2001), where flowering plant intrinsic rate of increase changed as a function of total flowering plant abundance. Here, responses were nonlinear but similar in form, as competition was limiting at high abundance while pollinator visitations were perhaps limiting where total flowering plant abundance was low (Ishii & Masahiko, 2001; Reilly et al., 2020). (d) In this case, the first derivatives of the intrinsic rates of increase varied as a function of flowering plant abundance, but did not differ much among species. (g) This resulted in a dissimilarity value close to its minimum (1) but that varied a little with total flowering plant abundance. (b) Next consider a case (based on Ford et al., 2018), where seedling establishment of several plant species responded nonlinearly to grazing pressure. This might be the case if, for example, one species is a shade-tolerant understorey specialist [light blue] that does well when grazing pressure is low but not when high, while another species is a shade-intolerant disturbance specialist [dark blue] (Ford et al., 2018). (e) These divergent nonlinear responses produced a range of first derivatives that changed as a function of grazing pressure, in turn producing (g) grazing pressure-dependent response diversity which was on, average, high in terms of dissimilarity, and was zero-spanning at intermediate grazing pressure. (c) Finally, consider a case (based on Altomare et al., 2021), where the density of several tree species responded nonlinearly to fire frequency with different functional forms. This might occur if some species were highly sensitive to burning [brown line] while others were disturbance specialists that require nutrient inputs from fire or reduced competition from dominant fire-sensitive species (Altomare et al., 2021). (f) Here, the first derivatives of tree density differed considerably and varied as a function of fire frequency, producing (i) generally high response diversity—in both dissimilarity and divergence—that also varied with fire frequency. Here, response diversity was largest at low and high fire frequency (when specialists responded strongly) and was lowest at intermediate fire frequency. Dashed lines indicate zero in panels a–f, and the median value of the similarity-based diversity metric (dissimilarity) along the X-axis in panels g–i. Note that as average divergence increases from g to i, the Pearson correlation coefficient (r) between dissimilarity and divergence increases: (g) Pearson's $r = 0.36$, (h) $r = 0.70$, (i) $r = 0.75$.

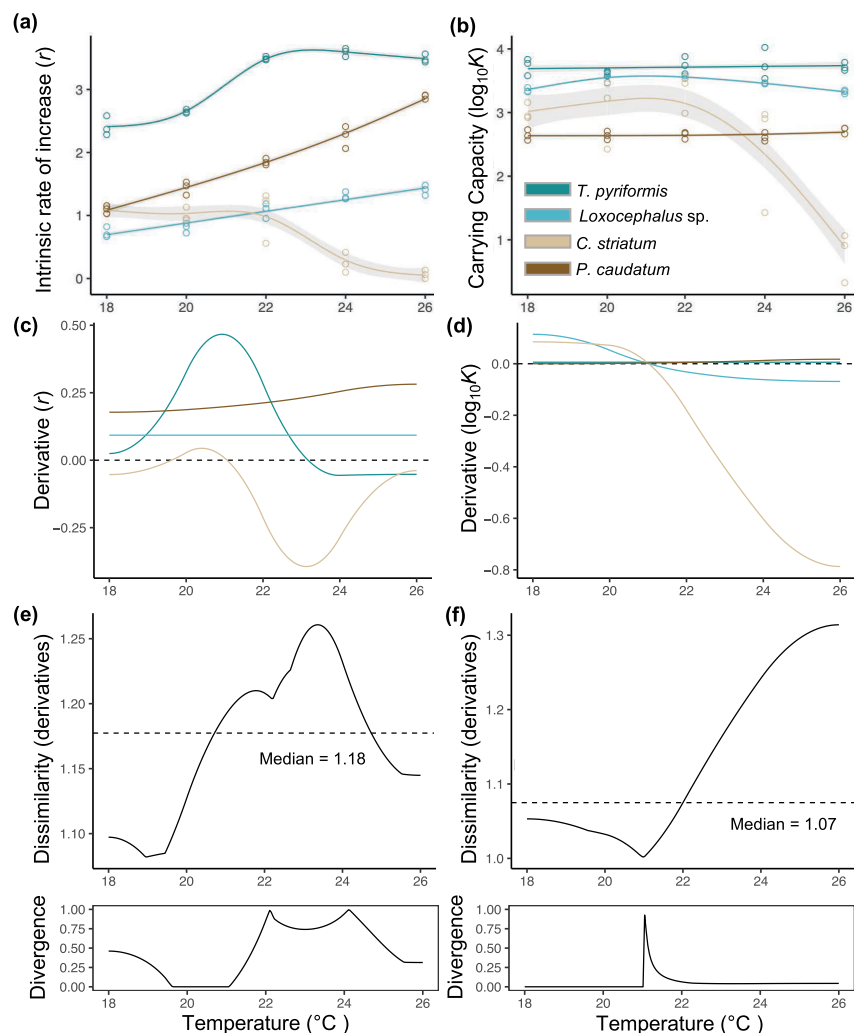


FIGURE 4 Reanalysis of data from Leary and Petchey (2009) using the framework proposed herein. Using empirical data from Leary and Petchey (2009), we measured performance–environment relationships for four aquatic ciliate species [coloured lines] as (a) the intrinsic rate of increase [r] and (b) the carrying capacity [$\log_{10}K$] along a temperature gradient (18–26°C). Data points are from a single population (see Figures 1a and 2a in Leary & Petchey, 2009). (c,d) We calculated the first derivatives of the performance–temperature relationships, which changed as a function of temperature due to nonlinear responses. (e,f) We then measured the similarity-based diversity and divergence of the derivatives of all species in the community for both (e) r and (f) $\log_{10}K$. (e) For r we found that both indices peaked around 22–24°C. (f) For $\log_{10}K$, dissimilarity peaked at the upper temperature limit, as the response of one species (*C. striatum*) diverged from the others (b,d), while divergence was mostly zero across the temperature range. Dashed lines indicate zero in panels c and d, and the median value of the similarity-based diversity metric (dissimilarity) along the X-axis in panels e and f.

divergence that also vary with the environment (Figure 3g–i). Using these hypothetical examples, we see that the performance variables of interest should be stable only over a narrow range of environmental conditions in the case of flowering plant abundance (Figure 3g), whereas seedling establishment and tree density are predicted to be stable over a wider set of environmental conditions (Figure 3h,i). In cases where there is no need to treat response diversity as environment dependent, one could estimate a single response diversity value across the environmental axis using any summary statistic (see dissimilarity in Figures 3g–i), although it may be appropriate to weight such a summary statistic by the distribution of the environmental variable. In linear cases, response diversity does not vary with environmental context, so summary statistics are not needed (Section 3.1).

By modelling individual species responses to environmental conditions with GAMs, this approach is conceptually similar to the binary species-specific interaction term described above (Cariveau et al., 2013; Winfree & Kremen, 2009; Figure 1). However, the framework we describe provides a quantitative estimate of response diversity, identifying precisely the relevant differences in the form of species environmental responses (dissimilarity and divergence)

that should drive ecological stability as described under the insurance hypothesis (Yachi & Loreau, 1999), and allowing consideration of response diversity as a function of the environment. This method is flexible enough to be used on traits, rates (first derivatives) or sensitivities of rates (second derivatives). We demonstrate cases taking first derivatives of traits before calculating response diversity (e.g. Figure 2b), thus producing rates from which we calculate response diversity (Figure 2e). However, it is also possible to estimate derivatives of rates (e.g. Figures 3d and 4), producing a measure of sensitivity of those rates to environmental change, which can be ecologically meaningful when measured over environmental gradients (Shyu & Caswell, 2014). However, study designs often estimate intrinsic rates of increase, for instance, under only two environmental contexts, such as in disturbed versus intact habitats. In such cases, it may be more meaningful to estimate response diversity based directly on the intrinsic rates of increase (rather than transforming them to second derivatives of abundance). Individual use cases will differ, so we suggest researchers carefully consider whether response diversity should be measured based on traits, rates or sensitivities, depending on available data and study goals (Figure 1). Practical users of response diversity can make use of the *mgcv* and *gratia* packages in R

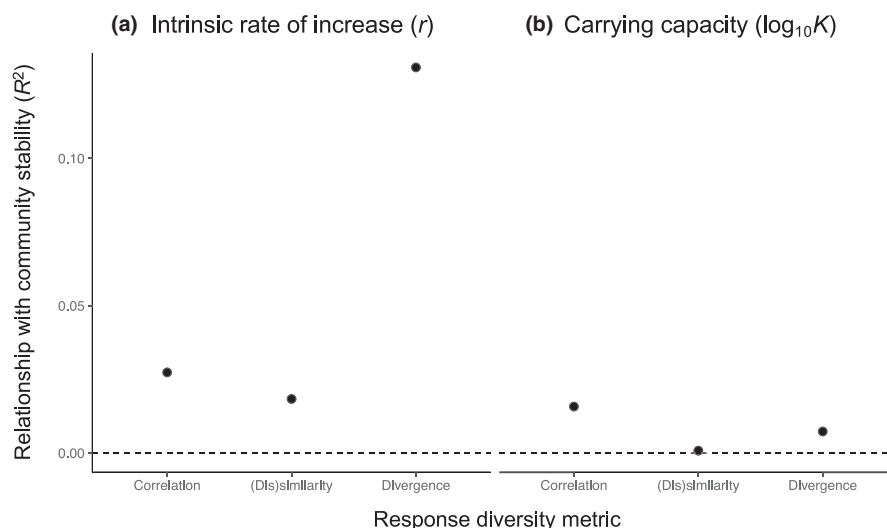


FIGURE 5 Comparison of response diversity–stability relationships for different metrics for data from Leary and Petchey (2009). After measuring response diversity for Leary and Petchey's (2009) ciliate dataset using the framework proposed herein, we compared the ability of response diversity to predict community stability (the coefficient of variation of total biomass for each pairwise combination of ciliate species; see Leary & Petchey, 2009; Ross, Petchey, et al., 2022b). Relationship with community stability is approximated by the predictive power (i.e. goodness-of-fit) using R^2 of the response diversity–stability relationship, where higher R^2 values represent higher predictive power. We tested three measures of response diversity: (1) a pairwise correlation-based measure similar to Leary and Petchey (2009) (Correlation on X-axis); (2) the Leinster and Cobbold (2012) similarity-based measure described therein ((Dis)similarity; Ross, Petchey, et al., 2022b); and (3) the divergence metric measuring the proportion of zero-spanning species' derivatives (Divergence). Panel (a) represents response diversity and community stability measured from the intrinsic rate of increase, while (b) is measured from the $\log_{10}$ carrying capacity (Leary & Petchey, 2009; Ross, Petchey, et al., 2022b). See Ross, Petchey, et al., 2022b for figures showing individual response diversity–stability relationships and a complementary R^2 figure relating response diversity to stability calculated using covariance rather than coefficient of variation. Note that while R^2 was highest for the divergence metric for the intrinsic rate of increase (a), none of the R^2 values here were particularly high, suggesting that even our best performing index did not predict community stability especially well.

to fit GAMs and calculate derivatives, respectively (Simpson, 2021; Wood et al., 2016), and we provide R code to calculate dissimilarity and divergence for the simulated case studies from Figures 2 and 3 and the reanalysis presented below (available through Github: Ross, Petchey, et al., 2022b).

Using the framework for measuring response diversity we present here, one can measure response diversity empirically in the laboratory or field. A conceptually simple and empirically tractable metric of response diversity can stimulate research on a range of previously unanswerable questions that can be experimentally tested in controlled systems. For example, once the range of response trajectories among species—or individuals/populations for cases of intraspecific response diversity—is established, communities encompassing differing amounts of response diversity can be manually assembled (Baskett et al., 2014; Leary & Petchey, 2009). In turn, this allows testing a variety of questions with response diversity (either dissimilarity or divergence) as the predictor variable, such as whether experimentally assembled communities with higher response diversity are more stable in terms of, for example, the temporal variability of total community biomass. Such stability analyses will require repeated temporal sampling of aggregate community biomass or ecosystem functioning. This then necessitates a decision about when to measure response diversity if analysing its relationship with temporal stability (see discussion in Box 1).

3.3 | Revisiting the response diversity–stability relationship in experimental aquatic communities

Next, as an empirical proof-of-concept for our framework, we demonstrate how it can be used to measure response diversity among aquatic microorganisms first presented by Leary and Petchey (2009). The original study involved four ciliate species (*Colpidium striatum*, *Loxocephallus* sp., *Paramecium caudatum* and *Tetrahymena pyriformis*). The first empirical step was to measure the responses of each species alone to temperature (five levels: 18, 20, 22, 24 and 26°C; Figure 4a,b). Responses of two types of performance were estimated from population time series: intrinsic rate of increase (r) and carrying capacity ($\log_{10}K$). These two properties (r and $\log_{10}K$) are uncorrelated here (Spearman's $\rho = 0.129$) and provide complementary information on species' performance, each with relevance at different temporal scales. For example, r is likely of more relevance if the environmental change of interest occurs rapidly or is short-lived, since r provides information on a population's ability to rapidly bounce back after disturbance. Conversely, $\log_{10}K$ may be most relevant over longer temporal scales of environmental change, since populations are unlikely to reach carrying capacity immediately following disturbance. The second empirical step was to record the biomass dynamics of communities that contained different combinations of the four species, and which were exposed to different

ranges of temporal variation in temperature. The original analyses involved calculating the response diversity present in each of the communities, and comparing this to the stability of the biomass dynamics of the communities (Leary & Petchey, 2009).

Here, we took a different approach to measure response diversity than in the original study. We modelled each species' performance (r or $\log_{10}K$) individually using GAMs and calculated the first derivatives of each performance-temperature relationship (Figure 4c,d). We then estimated the two dimensions of response diversity described above: the (dis)similarity of first derivatives (Leinster & Cobbold, 2012) and their divergence (the extent to which derivatives span zero for a given temperature). In doing this as a function of temperature, we produced temperature-dependent estimates of dissimilarity and divergence, which were highest for r at around 23.5°C when performance-environment responses were divergent and zero-spanning (Figure 4e), and highest for $\log_{10}K$ at the maximum temperature as the carrying capacity of *C. striatum* declines while the other species remain fairly constant (suggesting *C. striatum* has a disproportionate contribution to dissimilarity here). Note, however, that the divergence of the derivatives is low at higher temperatures for $\log_{10}K$, so we might predict any community-level properties driven by carrying capacity to be unstable at high temperatures (Figure 4f). Such temperature-dependent response diversity values provide an added layer of information—and hence predictive capacity—compared to the original analyses based on correlated performance-environment responses (Leary & Petchey, 2009).

Just as metrics measuring functional diversity with effect traits should be related to ecosystem functioning (Cadotte et al., 2011), response diversity measures should explain and/or predict community stability. Next, we assess this explanatory power, again using Leary and Petchey's (2009) aquatic ciliate experiment. To do so, we compared the relationship between response diversity and community stability of pairs of ciliate species. Stability is defined here as the coefficient of variation (ratio of the standard deviation to the mean) of aggregate ciliate biomass for each species pair (Donohue et al., 2013).

We calculated six measures of response diversity—three metrics, each applied to the r and $\log_{10}K$ response curves. We calculated the two metrics of response diversity discussed above: the Leinster and Cobbold (2012) similarity-based measure, and the divergence measure aimed at capturing whether species responses differ in their direction. We also calculated response diversity following the original study; we measured the correlation between the r or $\log_{10}K$ responses estimated for the species in each community. To assess the performance of different measures of response diversity, we then modelled the relationship between each response diversity metric and community stability using a simple linear regression. Specifically, for each species pair (see Leary & Petchey, 2009), we used Pearson correlations between each response diversity metric separately and the coefficient of variation of aggregate (two species) community biomass (using summed covariances as the stability measure produced similar results; Ross, Petchey, et al., 2022b). Where model slopes ($\pm 95\%$ confidence intervals [C.I.]) did not span zero, we interpreted

this as a significant relationship (positive or negative) with stability (Figure 5). This approach, based on the regression models from Leary and Petchey (2009) provides only a first and illustrative test of the performance of each response diversity metric. More sophisticated models of the relationship between response diversity and stability could likely account for some dependencies among observations, or for nonlinear relationships between response diversity and stability, which need not be simple linear functions (Ives et al., 1999). To concretely demonstrate that a given response diversity metric performs better than any other thus likely requires a priori information on the expected form of the response diversity-stability relationship. Using simulations to test metric performance for known response diversity-stability relationships will likely be an important step in validating new metrics. Nevertheless, our approach based on simple linear regression provides a first proof-of-concept for linking response diversity measured from performance-environment relationships to ecological stability and assessing the relative performance of different response diversity metrics.

For the intrinsic rate of increase, our proposed divergence metric outperformed other metrics of response diversity (including that from the original study) in terms of the predictive power (R^2) between response diversity and aggregate community biomass stability (Figure 5a). This metric, therefore, represents an advance when aiming to predict community stability of r , although the Leinster and Cobbold (2012) method was not a significant improvement, and divergence did not improve the relationship with $\log_{10}K$ stability (results of covariance-based stability were again similar; Ross, Petchey, et al., 2022b). Future efforts should thus use this example as a springboard from which to explore a wider range of response diversity methods, using data from various case studies, with the explicit goal of benchmarking any proposed new response diversity metric against existing approaches to determine which method best predicts community stability. In doing so, methodological development should not deviate from response diversity's a priori theoretical aim to reflect the diversity of species responses to the environment and, in turn, to mechanistically drive ecological stability via asynchrony (Yachi & Loreau, 1999).

4 | FUTURE DIRECTIONS AND CHALLENGES

The framework for measuring response diversity introduced above is conceptually simple and applicable across contexts. Using this approach, response diversity can be easily measured in empirical settings (see Figure 4 for reanalysis of data from Leary & Petchey, 2009), experimentally manipulated and related to stability where suitable community time series are available (Figure 5). This offers advantages over the functional trait approach dominant in the literature which estimates species' environmental responses less directly (Box 1 and Figure 1)—although future work should make direct comparisons between response diversity measured using performance-environment relationships and those estimated using low-level functional traits.

4.1 | Interspecific interactions

The performance–environment relationships we mention above can be established in isolation and then community-level response diversity calculated from individual species responses. However, this approach neglects any interspecific interactions within multispecies communities. Indeed, response diversity theory assumes that species–species relationships are less important for stability than abiotic environmental effects on species (i.e. species–environment relationships; Ives & Carpenter, 2007). Yet this assumption is not a limitation for two main reasons. First, one could simply model species responses (individually, using species-specific GAMs; Ross, Petchey, et al., 2022b) within a multispecies community, thereby allowing species–environment relationships to interact with species–species relationships as they would in nature (Armitage & Jones, 2020; Ives & Carpenter, 2007). Second, theoretical and empirical studies show that species–environment relationships are often more important than species–species relationships in driving both species asynchrony (e.g. Thibaut et al., 2012) and stability (e.g. Houlahan et al., 2007; Ives & Carpenter, 2007). That interspecific interactions need not be included when measuring response diversity is therefore not necessarily a limiting assumption, but rather a realistic one. That said, future work should establish the relevance of different interaction types when empirically measuring response diversity (Windsor et al., 2023).

4.2 | Multifarious environmental change

The framework above focuses on derivation of environment-dependent performance responses when a single environmental variable is varying. Yet, species and communities are often perturbed by multiple environmental stressors simultaneously. Rather than using response diversity solely as a predictor variable for estimating univariate stability responses, a system's response along an environmental gradient could be measured as a function of a second environmental condition. That is, response diversity can be assessed for a given environmental state and compared to other treatments or replicates across any environmental ranges of interest, permitting quantification of response diversity to multiple environmental stressors, which often occur simultaneously (Bowler et al., 2020; Dieleman et al., 2012). Another approach, which we aim to develop, would be based on multivariate response surfaces (e.g. Yang et al., 2022). With this, multivariate response diversity may link conceptually to the study of co-tolerance to multiple stressors and stress-induced community sensitivity (see Vinebrooke et al., 2004) and should prove more operationalisable than univariate approaches given the propensity for multiple environmental stressors to not only co-occur but also to interact non-additively in nature (Dieleman et al., 2012). Similarly, multivariate response diversity metrics that simultaneously capture two or more performance responses to a given environmental driver could be developed. Where such responses involve several ecosystem functions, this should inform on

the stability of ecosystem multifunctionality (Manning et al., 2018; Sasaki et al., 2019).

4.3 | Abundance and scale dependence

Response diversity depends on the identities and relative abundances of species within the community or functional group of interest. If community composition or abundance distributions change as a function of the environment, then response diversity should be abundance weighted to account for species losses and gains along the environmental gradient—for example, by changing the parameter q involved in the calculation of the similarity-based metric used here (Leinster & Cobbold, 2012). Such changes in species evenness can have knock-on consequences for species asynchrony (Thibaut & Connolly, 2013), and stability (Hillebrand et al., 2008). Accounting for evenness in response diversity measures is therefore needed to capture how evenness effects scale from species responses to community or ecosystem stability. Moreover, species identities and abundances vary as a function of scale, making response diversity scale dependent. Practical users of response diversity should be careful not to extrapolate results beyond the focal temporal or spatial scale given that ecological stability and its drivers vary in time and space (Jarrillo et al., 2022; Ross, Suzuki, et al., 2021b). Disturbance also has a temporal and spatial signature, meaning that the spatiotemporal scale of reference determines the environmental conditions and potential disturbances to which ecological communities are subject (Jackson et al., 2021; Zelnik et al., 2018). Thus, response diversity measured over short timescales does not represent a holistic picture of how communities are likely to respond to the environment (i.e. response diversity's goal to mechanistically capture the insurance capacity of a system), since the range of environmental conditions experienced at shorter scales may be limited when compared to longer temporal windows (Ross, Suzuki, et al., 2021b; Wolkovich et al., 2014; Zelnik et al., 2018). Despite the context dependencies outlined here, response diversity has the potential to be a powerful tool for predicting ecological stability. Future studies should determine the temporal or spatial scales over which specific metrics of response diversity (e.g. dissimilarity, divergence, response trait diversity) perform best when aiming to predict ecological stability and whether the methods presented here require further refinement to better reflect the insurance capacity of ecological communities across scales.

4.4 | Individual species contributions

Another extension of this framework would consider the relative importance of each species to community-level response diversity by, for example, measuring the median dissimilarity in performance–environment model slopes between each species and all others in the community. In this case, the species with the greatest dissimilarity from the others should be the largest contributor to

similarity-based response diversity and divergence (e.g. *Colpidium striatum* in Figure 4d), and thus be the species whose loss from the community should most greatly reduce response diversity and erode stability (provided the two are mechanistically linked). Such an approach complements studies considering the contribution of individual species to ecological responses and stability (e.g. Donohue et al., 2017; Ross, Arnoldi, et al., 2021a; Ross, García Molinos, et al., 2022a; White et al., 2020), in turn, aiding progress towards targeted conservation management decisions based on the quantifiable importance of individual taxa. Likewise, relative abundance and evenness are important when considering the contribution of species to response diversity, asynchrony and stability (Thibaut & Connolly, 2013). The similarity-based diversity metric we used here is based on Hill numbers, with an initial value of $q = 0$ indicating a focus on species richness. By increasing q to higher values, it should be straightforward to also incorporate relative species abundances (Leinster & Cobbold, 2012), and hence the potential role of dominance or rarity in shaping response diversity (e.g. Sasaki & Lauenroth, 2011). Importantly, the subjects of response diversity analyses need not be different species; individuals can also exhibit diversity in their response traits, further underscoring the importance of intraspecific variation in ecological responses to the environment (Barabás & D'Andrea, 2016; Herrando-Pérez et al., 2019; Mimura et al., 2017; Ross et al., 2017).

4.5 | Data resolution and spatiotemporal scope

Observational time series can also contain rich information on, for instance, community dynamics or environmental responses (Ives et al., 2003; Shoemaker et al., 2022; Ushio et al., 2018). Although methods for inferring causality from time series are emerging (e.g. *empirical dynamic modelling*, see Ross, Suzuki, et al., 2021b; Ushio et al., 2018), such methods are often complex, and results must be interpreted carefully. The same is true if measuring response diversity from observational time series; although response diversity may be a key biotic mechanism underlying the insurance effect of biodiversity (Yachi & Loreau, 1999), observational time series do not afford us information on causal relationships (but see Ushio et al., 2018). For dissimilarity, divergence or other metrics of response diversity to be measurable from observational data, time series should comprise higher-level traits (*sensu* Agrawal et al., 2010) as described above (e.g. measuring intrinsic—i.e. unlimited by competition—rather than realised rate of increase). Moreover, a high sampling resolution along environmental gradients is required to accurately differentiate between linear and nonlinear performance–environment relationships; longer or higher-resolution time series are better able to reveal nonlinear responses, and response diversity experiments should be designed with sampling resolution in mind to sufficiently capture all possible forms of performance–environment relationships. Compared to controlled experiments, observational data from the field are often ‘noisy’ and observation error can bias estimates of ecological interactions and response diversity (Ives et al., 2003),

requiring additional care when measuring response diversity. As well as ensuring appropriate performance variables (e.g. vital rates) are measured that capture some aspect of species' fitness or ecosystem functioning, one should aim to account for as many potentially confounding variables as possible when relating response diversity to stability, and to bear in mind that field studies rarely include all species from the regional pool, meaning response diversity is unlikely to portray the insurance capacity of the whole community. Future work should demonstrate the potential application of our response diversity framework to noisy field data. While the methods we describe here are useful for measuring stability from experimental or observational time series, dissimilarity and divergence could also be measured over spatial gradients to assess spatial variation in the stability and synchrony of functional properties such as primary productivity among habitat patches at a given spatial scale (Lamy et al., 2019; Loreau et al., 2003; Wang et al., 2019, 2021).

If response diversity can be measured from observational time series, it should be conceptually straightforward to harness global datasets (e.g. Dornelas et al., 2018) to measure response diversity at multiple sites across geographical regions, or to project locally measured response diversity to other locations. Although our response diversity framework produces results that are necessarily a function of the environmental conditions against which species' performances are assessed, comparing response diversity to relevant local environmental stressors across sites is informative insofar as it should predict stability. A global-scale analysis of response diversity to locally relevant environmental conditions should therefore unveil macroecological patterns and hotspots of ecological stability (Capdevila et al., 2022).

4.6 | Environmental dependencies of response diversity and stability

That response diversity can be environment dependent in cases where species respond nonlinearly to the environment has considerable implications for empirical studies of response diversity and for our understanding of ecological stability. Although threshold responses may be rare or difficult to detect in experiments (Hillebrand et al., 2020), nonlinear responses to the environment are common, and include functional responses to prey availability (Daugaard et al., 2019; Holling, 1966), and physiological responses to light and temperature (e.g. Payne et al., 2016; Peek et al., 2002). Nonlinear responses to the environment may result in environment dependencies in species asynchrony (Ghosh et al., 2020; Shoemaker et al., 2022). In turn, these can lead to environmentally dependent ecosystem multifunctionality and stability (Morin et al., 2014; Sasaki et al., 2019). Environment-dependent response diversity may therefore explain inconsistencies in the reported effects of biodiversity on ecological stability across studies (Jacquet et al., 2016; Pennekamp et al., 2018). Importantly, the empirical framework we propose permits identification of the specific environmental conditions under which response diversity should or should not be stabilising. For a nonlinear environmental response, $f(E)$,

one can examine the values of E (the 'environment') for which the derivatives of performance–environment relationships, $f'(E)$, including their confidence regions, diverge (span zero). In other words, our framework makes it conceptually and practically simple to identify a priori hypotheses regarding the stabilising effect of response diversity based on environmental conditions. This should complement existing studies of the impact of environmental conditions on ecological stability (e.g.; Hillebrand et al., 2018; Ross, Suzuki, et al., 2021b) by identifying possible underlying biotic drivers of environmental responses. Moreover, in cases where response diversity varies as a function of the environment (i.e. where performance–environment relationships are nonlinear), it is possible to manipulate response diversity without altering community composition. Comparing ecological stability between identical communities under different (fixed) environmental conditions, might then allow identification of response diversity's mechanistic stabilising role. By controlling for species identity in this way, there are fewer potential confounding factors driving any observed community-level responses (although environmental context is added as an extrinsic driver). For example, one could measure the diversity–stability relationship for a plant community with known performance responses to drought under drought and nondrought conditions to mechanistically consider how drought relates to stability. There is still much to learn about context-dependent ecological stability as it relates to nonlinear performance–environment relationships and environment-dependent response diversity, and we suggest this as a promising avenue of future work.

5 | CONCLUSION

Since the observation that diversity alone is insufficient to explain the stabilising effect of species richness on aggregate ecosystem properties (Ives et al., 1999; May, 1973), ecologists have been searching for a general mechanism explaining the insurance effect of biodiversity across contexts (Yachi & Loreau, 1999). Here we proposed response diversity as a promising but understudied mechanism for predicting ecological responses to environmental conditions (Elmqvist et al., 2003; Mori et al., 2013; Nyström, 2006). We found that empirical studies of response diversity were rare, and where they did occur, most measured the diversity of response traits (e.g. Laliberté et al., 2010; Morel et al., 2020). However, given the challenges associated with reliably producing response diversity estimates from low-level trait data (Bartomeus et al., 2017; Box 1), we suggest measuring response diversity more directly as the variation in performance responses to environmental conditions (Leary & Petchey, 2009). The framework we proposed here provides a direct, empirically tractable and conceptually simple estimate of response diversity based on dissimilarity and divergence in performance–environment relationships that should facilitate robust empirical tests of the diversity–stability relationship across taxa and environments. Such tests have the potential to identify the long sought-after mechanisms underpinning biodiversity's role in buffering ecosystems against environmental change.

AUTHOR CONTRIBUTIONS

Samuel R.P.-J. Ross was involved in conceptualisation, software, formal analysis, funding acquisition, validation, investigation, visualisation, methodology, writing—original draft and writing—reviewing and editing. Owen L. Petchey was involved in conceptualisation, data curation, software, validation, methodology and writing—reviewing and editing. Takehiro Sasaki was involved in conceptualisation, methodology and writing—reviewing and editing. David W. Armitage was involved in conceptualisation, software, formal analysis, methodology and writing—reviewing and editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

PEER REVIEW

The peer review history for this article is available at <https://www.webofscience.com/api/gateway/wos/peer-review/10.1111/2041-210X.14087>.

DATA AVAILABILITY STATEMENT

All data and R code necessary to reproduce the analyses and figures are available in the Zenodo Digital Repository <https://doi.org/10.5281/zenodo.7018707> (Ross, Petchey, et al., 2022b).

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SUPPORTING INFORMATION

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Figure S1. PRISMA diagram.

List of papers included in final results after screening for eligibility.

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